DEEPAK KATUKURI

FULL-STACK PRODUCT ENGINEER · APPLIED AI



CONTACT

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TECHNICAL SKILLS

Languages: TypeScript / JavaScript, Python, SQL, PL/SQL, Java, C++

Frontend: React, Next.js-ready, Vite, Tailwind CSS, React Query, Recharts, component / design systems, data-dense dashboards, empty / loading / error states, responsive & dark-mode UI

Backend & APIs: FastAPI, REST APIs (auth, status codes), Node / TypeScript services, PostgreSQL / Supabase-style Postgres, schema design & migrations, background jobs, Docker

Applied AI / LLM: RAG, retrieval, embeddings, vector search, document processing (text + visuals), prompt engineering, CLIP, DistilBERT, Ollama / OpenAI- / Anthropic- / Gemini-compatible APIs

Dev & Cloud: Git / GitHub, PR workflows, code review, CI/CD, AI-assisted dev (Cursor / Claude Code), cloud & object storage, pytest

ML / Deep Learning: PyTorch, TensorFlow, scikit-learn, XGBoost, CNNs, fine-tuning, NumPy, Pandas

EDUCATION

MSc Data Science
University of Europe, Germany
Sep 2025 – Present

BTech – CS & Engineering
MLRIT, India
Jun 2018 – Jul 2022

CERTIFICATIONS

PL/SQL – Database Management
Core Java – Software Development
OneShield – P&C Insurance

LANGUAGES

English – Fluent
German – Beginner (A1/A2)

PROFILE

Full-stack product engineer and Data Science MSc candidate who ships features end-to-end — frontend, backend, and the AI behind them — and puts them in front of real users. Built a full-stack product with a React / Tailwind frontend on a Python + PostgreSQL backend, plus retrieval and document-processing LLM systems written from scratch (no framework wrappers). Comfortable in TypeScript / JavaScript and React / Tailwind, backed by 2.5 years of production engineering — I knew how to code before Cursor and Claude Code and use them to move faster, not to think for me. Drawn to acemate's mission of building AI that helps educators teach at scale, and eager to own product from user problem to deployment.

PROJECTS

RA Nutrition & Lifestyle App — Full-Stack AI-Powered Product

React · Vite · Tailwind CSS · React Query · Recharts · FastAPI · PostgreSQL

- Designed and built a health product end-to-end — React / Tailwind frontend on a FastAPI + PostgreSQL backend — with JWT authentication + token refresh, meal / symptom logging, and a transparent rule-based recommendation engine.
- Built a data-dense weekly analytics dashboard (React Query + Recharts) that keeps user trends readable at a glance, with explicit empty, loading, and error states and a reusable Tailwind component system.
- Modeled normalized data across 4 logical PostgreSQL schemas (core, static, tracking, intelligence); architected a V3 ML tier (scikit-learn / XGBoost + PyTorch LSTM / GRU / TCN), gated to ship only where it beats the rule-based baseline.

RAG Application — Document Processing & Retrieval from Scratch

Python · Ollama / LLM APIs · sentence-transformers · github.com/Deepak589/RAG_pipeline

- Built a framework-free RAG application in ~270 lines of Python (no LangChain or vector DB) — document chunking, embedding, similarity search, and prompt assembly — a real AI feature, not a chatbot wrapper, degrading gracefully when offline.
- Cut repeat embedding cost with a SHA-256-fingerprinted vector cache that auto-invalidates on chunk / model change, pairing a dense bi-encoder retriever (all-MiniLM-L6-v2, cosine top-3) with a TF-IDF baseline; extending to hybrid RRF fusion + cross-encoder reranking.

Food Image-to-Recipe Retrieval — Multimodal AI (text + visuals)

PyTorch · CLIP ViT-B/32 · DistilBERT · InfoNCE · MSc final project

- Built a config-driven multimodal app end-to-end — a dual-encoder (frozen CLIP ViT-B/32 + DistilBERT) mapped into a shared embedding space via symmetric InfoNCE — reaching validation median rank 10/200 on 13.5K images and cutting training ~10x by caching CLIP features, with 58 unit tests. (0.98 ROC-AUC EfficientNetB0 CT-scan classifier also built.)

EXPERIENCE

Software Engineer ValueMomentum — *Mar 2022 – Sep 2024*

- Delivered features across 2 enterprise applications from concept to deployment on OneShield PAS, owning work end-to-end in an agile team with code reviews, sprint planning, and daily stand-ups.
- Built and optimized PL/SQL stored procedures and functions powering backend data operations and endorsement modules serving live policy processing.
- Accelerated releases by running QA / UAT test cases and supporting CI/CD deployment of fixes across the full policy lifecycle.

Machine Learning Intern SmartBridge × IBM — *Jun – Jul 2019*

- Built and deployed a lung-cancer risk-prediction ensemble (Random Forest, KNN, Gradient Boosting, Neural Network) via IBM Watson ML, communicating results with Matplotlib / Seaborn visualizations.